NAME: ______________________________________ CT: 10S_______

Centre Number: _____________________ Index Number: ____________

HWA CHONG INSTITUTION

C2 PRELIMINARY EXAMINATION
H1 CHEMISTRY 8872
PAPER 2

21 September 2011 **2 hr**

Candidates answer **Section A** on the Question Paper.
Additional Materials: Answer Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST
Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue, correction fluid or tapes.

Section A
Answer **all** the questions.

Section B
Answer **two** questions on separate answer paper.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINERS' USE ONLY

						TOTAL
Multiple Choice	Section A (Structured)		Section B (Free Response)			
	Q1	/ 12	Q5	/ 20		
	Q2	/ 11	Q6	/ 20		
	Q3	/ 10	Q7	/ 20		
	Q4	/ 7				
/ 30	Subtotal	/ 40	Subtotal	/ 40		110

Section A

Answer **all** the questions in this section in the spaces provided.

1 Calcium ethanedioate, CaC_2O_4, is a white needle-like crystalline solid. It can be prepared by reacting calcium hydroxide, $Ca(OH)_2$, and ethanedioic acid, $H_2C_2O_4$.

(a)(i) Name the reaction between calcium hydroxide, $Ca(OH)_2$, and ethanedioic acid, $H_2C_2O_4$.

Write an equation for the reaction involved.

....

....

(ii) State all the bonding types in CaC_2O_4.

....

....

[4]

(b) When a pure sample of anhydrous CaC_2O_4 was heated strongly at 400 °C until no further change in mass is observed, a white solid **B** and 0.028 g of carbon monoxide gas were obtained as the **only products**.

(i) Given that 1 mole of CaC_2O_4 decomposes to give 1 mole of CO(g), write a balanced equation for the thermal decomposition of CaC_2O_4 and identify solid **B**.

....

(ii) Hence, determine the mass of **B** obtained in the reaction.

[4]

(c) In an experiment, 25.0 cm^3 of 0.050 mol dm^{-3} $K_2C_2O_4(aq)$ was titrated against 0.022 mol dm^{-3} $KMnO_4$ in an acidic medium. It was found that 22.70 cm^3 of $KMnO_4(aq)$ was needed for the complete reaction, alongside with effervescence being observed during titration.

Given the reaction of manganate(VII) in acidic medium as:

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

(i) Calculate the mole ratio of $K_2C_2O_4$ to $KMnO_4$ in the reaction.

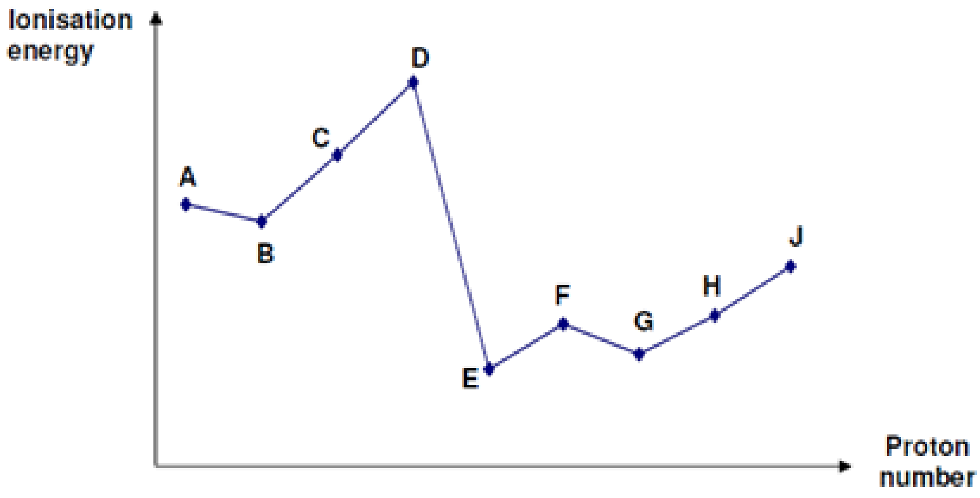
(ii) Hence, write the half-equation for the reaction of $K_2C_2O_4$ and the balanced equation for the reaction between $K_2C_2O_4$ and acidified $KMnO_4$.

....

....

[4]
[Total: 12]

2 The ionisation energies of nine consecutive elements **A** to **J** in Periods 2 and 3 of the Periodic Table are as shown.



(a) **A** and **J** are in the same group. Explain why the ionisation energy of **J** is lower than that of **A**.

....

....

[1]

(b) The oxide of **H** is insoluble in water while its chloride reacts completely with aqueous sodium hydroxide to give a resulting mixture of pH close to 7.

(i) To which group does **H** belong? ……………………………………………………….…

(ii) Using the symbol **H**, write the formula of the oxide of **H**. …………………………….

[2]

(c) One mole of the chloride of **G** reacts with one mole of the hydride of **A** to form a product.

In the boxes below, draw the shapes of the chloride of **G**, hydride of **A** and the product formed, indicating clearly the bond angles.

chloride of **G**	hydride of **A**	product formed

[4]

(d) **J** can form different chlorides of JCl_3 and JCl_5.

State the pH of the resultant mixture when the chloride of **G** and one of the chlorides of **J** is added to water separately. Write an equation for each to support your answer.

	Chloride of **G**	Chloride of **J**
Equation to show reaction with water (if any)		
pH of solution formed when added to water		

[4]
[Total: 11]

3 **K**, **L**, **M** and **N** are consecutive elements of increasing proton number in the Periodic Table.

(a) The successive ionisation energies, in kJ mol^{-1} of element **L** are given below.

1681 3374 6050 8407 11020 15160 17870 92040 106400

To which Group does element **L** belong? Explain your answer.

....

....

[2]

(b) **K**, **L** and **N** form ions that are isoelectronic with atom **M**.

(i) State the charge on the ions of **K**, **L** and **N**.

K: ………………… **L**: ………………… **N**: …………………..…

(ii) Arrange the ions of **K**, **L** and **N** in *decreasing* order of ionic radius. Explain your answers.

....

....

....

[3]

(c) Draw a dot-and-cross diagram each to show the bonding between:

(i) **K** and **L**	**(ii)** **K** and **N**.

[2]

(d) The compound in **c(ii)** has a high melting point.

(i) Explain, in terms of structure and bonding, why the melting point of this compound is much higher than element **N**.

....

....

....

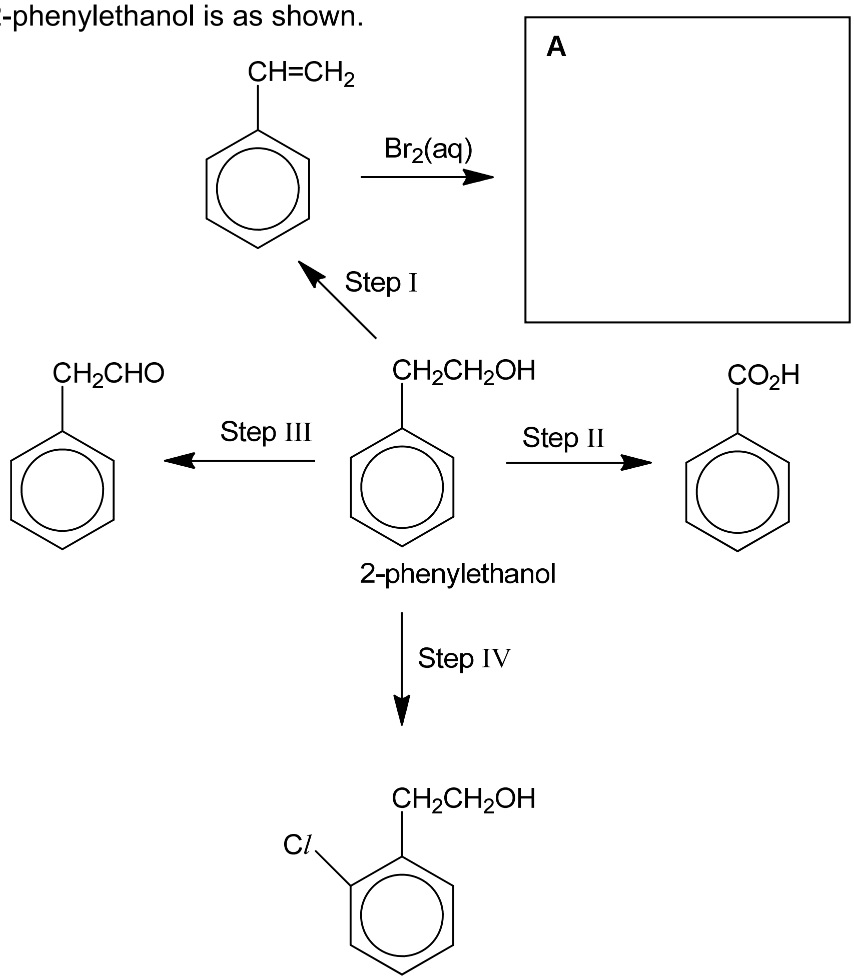
(ii) Do you expect the compound in **c(ii)** to conduct electricity in all states? Explain your answer.

....

....

[3]
[Total: 10]

4(a) 2-phenylethanol is one of the chemicals used as additives in cigarettes. A reaction scheme starting from 2-phenylethanol is as shown.



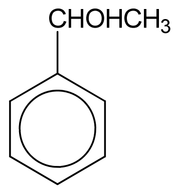
(i) Draw the structure for compound **A** in the box above.

(ii) State the reagents and conditions for Steps **I**, **II**, **III** and **IV**.

Step	reagents and conditions
I	
II	
III	
IV	

[5]

(b) 1-Phenylethanol is an isomer of 2-phenylethanol.



1-phenylethanol

Suggest a simple chemical test by which 1-phenylethanol can be distinguished from 2-phenylethanol. State the reagents and conditions and the observations expected for each.

	1-phenylethanol	2-phenylethanol
Reagents and conditions		
Observations		

[2]
[Total: 7]

SECTION B

Answer **two** of the three questions in this section on separate answer paper.

5(a) The enthalpy change of combustion of propan-2-ol is –2006 kJ mol^{-1}.

(i) Calculate the mass of propan-2-ol that needs to be burned to bring 1.00 kg of water at 30 °C to boiling. Assume that the process is 80% efficient and the specific heat capacity of water is 4.2 J g^{-1} K^{-1}.

(ii) Consider the following energy cycle involving propan-2-ol.

$$\begin{tikzcd}
CH_3CHOHCH_3(l) + \frac{9}{2}O_2(g) \arrow[rr, "\Delta H_2"] & & 3CO_2(g) + 4H_2O(l) \\
& C(s) + 4H_2(g) + 5O_2(g) \arrow[ul, "\Delta H_1"] \arrow[ur, "\Delta H_3"'] &
\end{tikzcd}$$

I Define enthalpy change of combustion of propan-2-ol.

II What does ΔH_1 represent?

III Using the following data, calculate ΔH_1.

enthalpy change of combustion of carbon = –393 kJ mol^{-1}

enthalpy change of combustion of hydrogen = –285 kJ mol^{-1}

enthalpy change of combustion of propan-2-ol = –2006 kJ mol^{-1}

[7]

(b) **P**, $C_6H_{12}O_2$, is a neutral compound. Heating **P** with aqueous acidified potassium manganate(VII) gives compounds **Q**, $C_3H_6O_2$, and **R**, C_3H_6O. Effervescence is observed when **Q** reacts with aqueous sodium carbonate. **R** reacts with 2,4-dinitrophenylhydrazine but not with Fehling's solution.

Deduce the structural formula of **P**, **Q** and **R**. Explain the reactions described and write equations where appropriate.

[7]

(c) Suggest the reagents and conditions for the following conversions. Give the structural formulae of all intermediate compounds formed.

(i) CH_3CH_2Br → **Q**, $C_3H_6O_2$

(ii) $CH_3CH_2COCH_3$ → **Q**, $C_3H_6O_2$

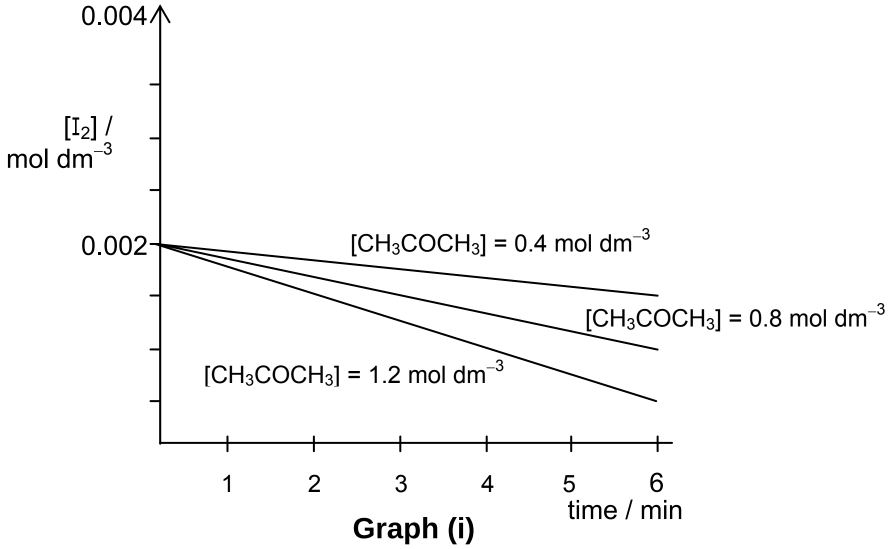
(iii) $CH_3CH{=}CH_2$ → **R**, C_3H_6O

[6]
[Total: 20]

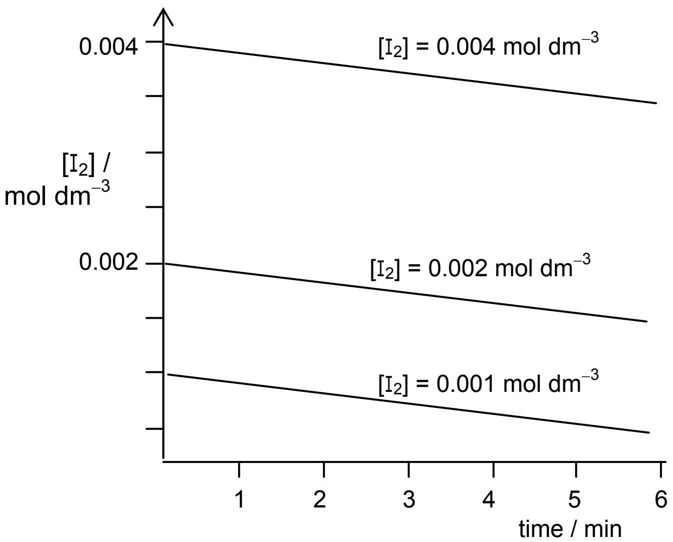
6(a) The kinetics of the acid-catalysed reaction of propanone with iodine can be investigated experimentally by varying the concentrations of the three substances involved.

$$CH_3COCH_3(aq) + I_2(aq) \xrightarrow{H^+} CH_2ICOCH_3(aq) + HI(aq)$$

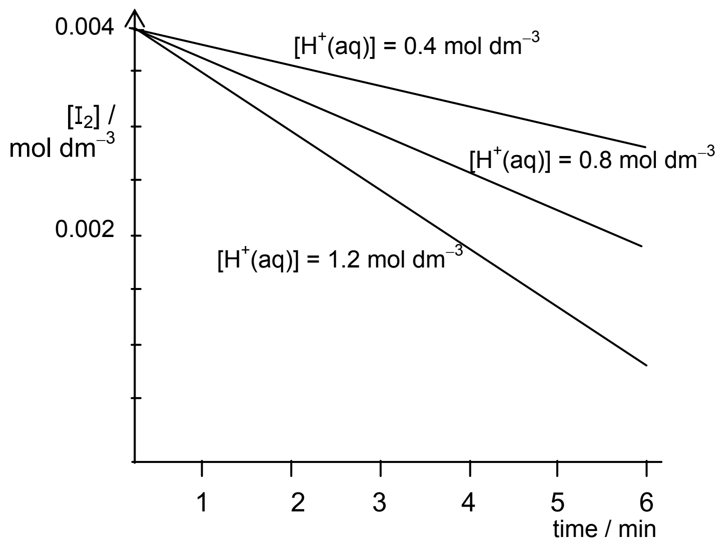
The following results were obtained in an experiment.



Graph (i)



Graph (ii)



Graph (iii)

(i) Deduce, with reasoning, the orders of reaction with respect to:

I I_2, **II** H^+, and **III** CH_3COCH_3.

(ii) Hence, write a rate equation for the reaction and determine the value for the rate constant *k*, including its units.

(iii) Predict, with a reason each, the effect of a catalyst on the activation energy and rate constant of a reaction.

[10]

(b) Geranyl acetate is a natural constituent of many essential oils and is used in perfumes and as flavorings.

$(CH_3)_2C{=}CHCH_2CH_2CCH_3{=}CHCH_2OCOCH_3$ geranyl acetate

(i) State the type of isomerism geranyl acetate displays. Draw the isomers clearly.

(ii) Describe in terms of '**overlap of orbitals**' in the formation of C=C bonds in geranyl acetate.

(iii) Draw the structural formulae of the products you would expect from the treatment of geranyl acetate with each of the following reagents. In each case, name the type of reaction that occurs.

I hydrogen bromide gas.

II hot aqueous sodium hydroxide.

[10]
[Total: 20]

7(a) The reaction in the Contact process for manufacturing sulfuric acid is as follow.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

When sulfur dioxide and oxygen are mixed, the two gases slowly react to form sulfur trioxide according to the equilibrium.

(i) Sketch a graph to show how the rates of the forward and reverse reactions change from the time the two gases are mixed to the time when the reaction reaches equilibrium. Label the two lines clearly.

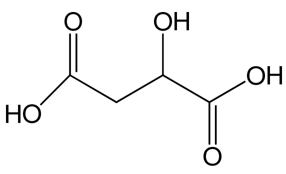
(ii) At a specific temperature, 2.0 moles of $SO_2(g)$ and 1.0 mole of $O_2(g)$ were added in a 2.0 dm^3 flask. The equilibrium mixture was found to contain 1.60 moles of $SO_3(g)$.

I Write the expression for the equilibrium constant, K_c.

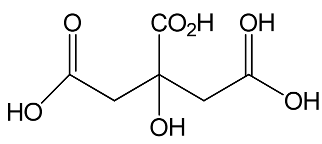
II Calculate the equilibrium concentrations of $SO_2(g)$, $O_2(g)$ and $SO_3(g)$ and hence the equilibrium constant, K_c, of the reaction at this temperature.

[8]

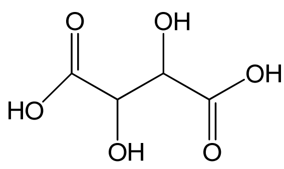
(b) The following are acids found in apples, oranges and grapes respectively.



malic acid



citric acid



tartaric acid

(i) Draw the structural formula of the organic product formed when PCl_5 is added to malic acid.

(ii) Describe a simple chemical test that can distinguish citric acid from the other two acids and describe what you would observe.

(iii) Show how malic acid can be converted to tartaric acid in the laboratory via **two** steps. In your answers, show clearly the reagents, conditions and intermediate products involved. Write equations for the reactions that occur.

[6]

(c) Malic acid, $C_3H_5O_3CO_2H(aq)$, in apple juice can be considered to be a weak monobasic acid that dissociates partially in water.

$$C_3H_5O_3CO_2H(aq) \rightleftharpoons C_3H_5O_3CO_2^-(aq) + H^+(aq)$$

(i) The pH of a sample of apple juice is 2.50. Calculate the molar concentration of hydrogen ions, $[H^+]$, in the juice.

(ii) In a titration reaction, 20.0 cm^3 of the apple juice was found to react completely with 23.30 cm^3 of 0.15 mol dm^{-3} NaOH(aq). Calculate the concentration of malic acid in the juice.

(iii) Explain the difference in the results obtained in parts **d(i)** and **d(ii)**.

(iv) Suggest, with a reason, a suitable indicator for the titration.

(v) The aqueous solution of malic acid and sodium malate constitutes a buffer system.

With the aid of suitable equations, explain how such a system works

I when small amounts of alkali are added.

II when small amounts of acid are added.

[6]
[Total: 20]

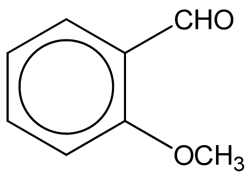
End of Paper 2

Extra

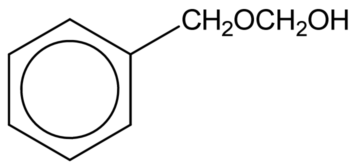
Organic compounds can be distinguished using simple chemical tests. Suggest how you can distinguish between the following pairs of isomers. State clearly the reagents and conditions used as well as the observations for each compound.
[The $–OCH_3$ group is inert and can be disregarded in this question.]

(i) $CCl_3CH_2COCH_3$ and $CCl_3COCH_2CH_3$

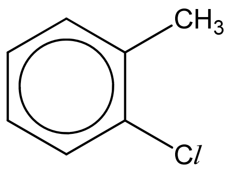
(ii)



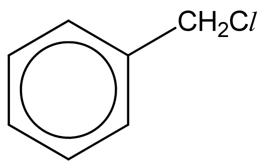
and



(iii)



and



(b)(i) When burnt in a limited supply of air, ethane forms carbon monoxide and water. The enthalpy change of the reaction is –1004 kJ mol^{-1} of ethane.
Construct a balanced equation that represents this enthalpy change.

(ii) Ethane can be synthesized through the hydrogenation of ethene as shown in the following equation:

$$CH_2{=}CH_2(g) + H_2(g) \rightarrow CH_3CH_3(g)$$

Calculate the enthalpy change of the hydrogenation reaction, using the following data in addition to the one given in **(b)(i)**.
Enthalpy change of combustion of $CH_2{=}CH_2(g)$ = –1411 kJ mol^{-1}
Enthalpy change of combustion of $H_2(g)$ = –286 kJ mol^{-1}
Enthalpy change of combustion of CO(g) = –283 kJ mol^{-1}

[4]

(c) Ethane can be used to synthesize propanoic acid as shown in the following reaction scheme:

$$CH_3CH_3 \xrightarrow{\text{I}} \textbf{P} \xrightarrow{\text{II}} \textbf{Q} \xrightarrow{\text{III}} CH_3CH_2CO_2H$$

propanoic acid

(i) Identify the intermediate compounds **P** and **Q**.

(ii) State the reagents and conditions for steps **I** to **III**.

[4]

(a) Coke contains phosphoric acid, H_3PO_4, a tribasic acid. In this part of the question, H_3PO_4 is considered as a weak *monobasic* acid that dissociates partially in water.

$$H_3PO_4(aq) \rightleftharpoons H_2PO_4^-(aq) + H^+(aq)$$

(i) The pH of a sample of coke is 3.20. Calculate the molar concentration of hydrogen ions, $[H^+]$ in the coke.

(ii) A sample of coke was treated with activated carbon to remove the colouring before titration with aqueous NaOH. It was found that 20.0 cm^3 of the coke reacted completely with 23.30 cm^3 of 0.15 mol dm^{-3} NaOH(aq).
Calculate the concentration of phosphoric acid in the coke.

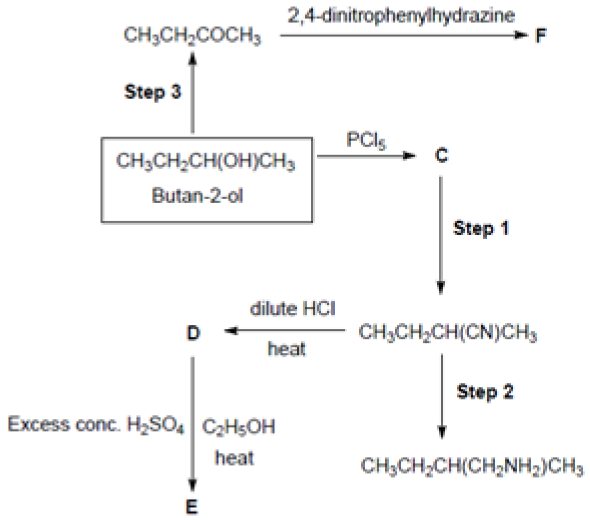
(iii) Explain the difference in the results obtained in parts **(i)** and **(ii)**.

(iv) Suggest, with a reason, a suitable indicator for the titration.

(v) Write an expression for the acid dissociation constant, K_a, of phosphoric acid.

(b) Phosphoric acid has multiple dissociation constants and can be used to prepare phosphate buffers at pH values of 2, 7 and 12. One such buffer contains $NaH_2PO_4$and Na_2HPO_4 and is used mainly in biological applications.
Explain, with relevant equations, how the system regulates the pH
(i) when small amounts of an alkali are added.
(ii) when small amounts of an acid are added.

(c) The diagram below shows a series of reactions starting from butan-2-ol.



(i) Draw the structural formulae for the compounds **C** to **F**. [4]

(ii) State the reagents and conditions for Steps **1** to **3**.
[3]

(iii) When butan-2-ol reacts with excess concentrated sulphuric acid, three possible alkenes, **G**, **H** and **J** are formed. The table below gives the products formed when **G**, **H** and **J** react with hot acidified KMnO4.

Alkene	Products formed
G	$CO_2(g)$ + CH_3CH_2COOH
H	2 moles of CH_3COOH
J	

Draw the structural formula of these three alkenes formed and state the type of isomerism exhibited by the alkenes **H** and **J**. [4]
[Total: 20]

2(a) Sodium peroxodisulfate, $Na_2S_2O_8$, is commonly used as a bleaching agent in hair cosmetics. Given that peroxodisulfate oxidises iodide solution to form black crystals, while it is reduced to form SO_4^{-2}, construct a balanced equation. [1]

A quality control experiment was set up to monitor the current level of peroxodisulfate in the air of the laboratory to ensure that it does not reach hazardous levels (above 1000 mg dm^{-3}).

25 cm^3 of air was bubbled into 250 cm^3 of water and 25 cm^3 of this solution was taken to titrate against a standard solution of potassium iodide, with known concentration of 0.0025 mol dm^{-3}.

(i) Assuming that all of the peroxodisulfate particles were dissolved into the solution, and that it took 10.25 cm^3 of iodide solution to fully react with the peroxodisulfate present, calculate the mass of peroxodisulfate in 1 dm^3 of air.

(ii) Comment on whether the laboratory needs to be temporarily closed down for disinfection. [3]

The reaction between peroxodisulfate and iodide is slow. However, when trace amounts of iron(II) nitrate was added, the reaction took place readily. With the aid of a relevant diagram, explain the observations above and state the role of iron(II) nitrate.

(b) Sodium peroxodisulfate, $Na_2S_2O_8$, undergoes redox reaction with potassium iodide, KI, and the rate of reaction was followed by measuring the change in concentration of $Na_2S_2O_8$, with time.

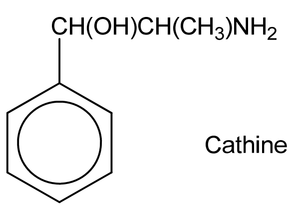
Two experiments were carried out starting with different concentrations of KI. The following results were obtained.

Time / minutes	Expt 1, with [KI] = 0.10 mol dm^{-3} [$Na_2S_2O_8$] / mol dm^{-3}	Expt 2, with [KI] = 0.50 mol dm^{-3} [$Na_2S_2O_8$] / mol dm^{-3}
0	0.0050	0.0050
15	0.0040	0.0045
30	0.0032	0.0040
45	0.0026	0.0036
60	0.0021	0.0032
75	0.0017	0.0029
90	0.0014	0.0026

(i) Using the same axes, plot the graphs of [$Na_2S_2O_8$] against time for the two experiments.
(ii) Use your graphs to determine the order of reaction with respect to [KI] and to [$Na_2S_2O_8$], showing your working clearly. Hence give the rate equation for the reaction.

Calculate the initial rate for experiment 1, and use it, together with your rate equation, to calculate the rate constant for the reaction, including units. [8]

5 Cathine is a stimulant found in a type of plant named Khat. Stimulants are drugs which can temporarily increase alertness and wakefulness. The World Anti-Doping Agency bars athletes from participating in international sports events, like the Olympics, if concentration of cathine is above 5.0 x 10^{-6} g cm^{-3} of urine.

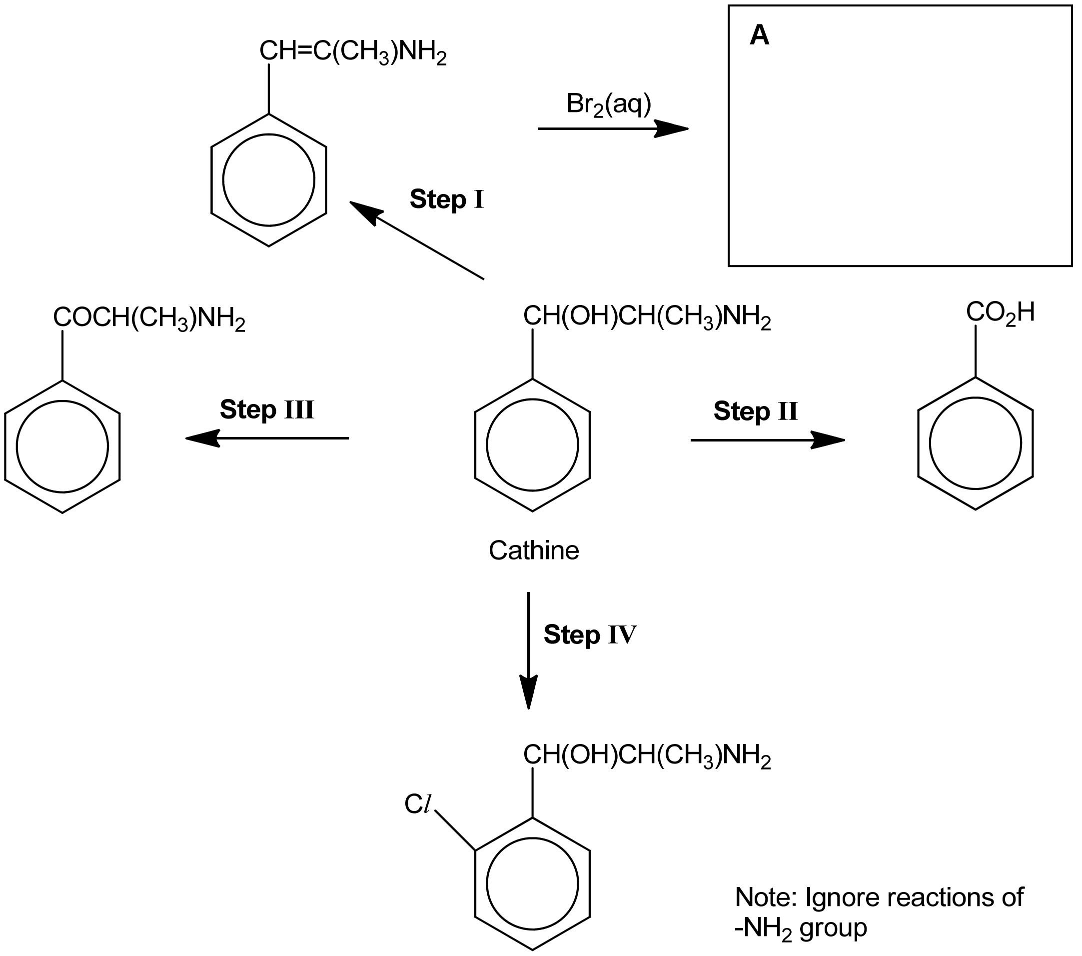


Cathine

(a) A sample of urine obtained from an athlete is 300 cm^3.
What is the minimum amount of Cathine in urine to reach the World Anti-Doping Agency maximum legal concentration of 5.0 x 10^{-6} g cm^{-3}?

[2]

A reaction scheme starting from Cathine is as shown.



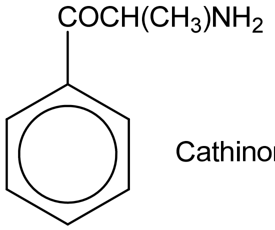
(i) Draw the structure for compound **A** in the box above.

(ii) State the reagents and conditions for **Steps I**, **II**, **III** and **IV**.

Step	reagents and conditions
I	
II	
III	
IV	

[5]

(c) Cathinone, a more powerful stimulant than Cathine, is also found in Khat.



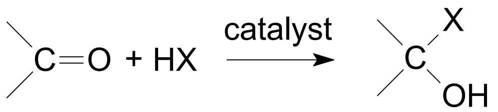
Cathinone

Suggest a simple chemical test by which Cathine can be distinguished from Cathinone. State the Reagents and conditions and the observations expected for each.

	Cathine	Cathinone
Reagents and conditions		
Observations		

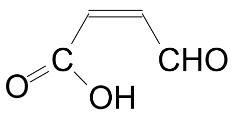
[2]

(b) The carbonyl functional group undergoes a series of reactions which follow the following pattern:



where X can be CN, OCH_3, etc.

Using the above information, draw the structural formula of the product formed for the following reactions:

(i) Addition of HCN to 

, followed by a dehydration process.